

FYP - OpenPaths

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1 Literature Review

Transportation and its different modes have overarching effects in a multitude of different sectors, namely the environment, health and socioeconomic relationships. Car centric development has lead to a rise in carbon emissions [?], costly road maintenance [?], and hostile public spaces. Shifting our cities into a walkable public and focusing on alternative transportation such as cycling, will help people lead an active lifestyle and allow for safe spaces. It is for these reasons that it is critical to create a tool to assist urban planners in shifting roads away from cars and reallocating them to cyclists. The aim is to expand a safe cycling network all through Malta reaching all major points of interest while maintaining a fully connected car network.

Before discussing the different approaches we must understand the fundamental similarities that all approaches take. The underlying road infrastructure is expressed as a graph network, where intersections are nodes and are connected by road segments, i.e edges. The network is typically a directed multi-graph, where each edge could only flow in one direction, and each node pair could have multiple edge connections, where each edge emulates a single lane within the road segment. Edge weights typically encode travel time, computed from distance, gradient and [1] use perceived safety for bikes, whilst cars use a simpler distance/speed encoding. Furthermore [2] make explicit use of multi layer networks where there are is the car road network and a separate protected cycle network.

The aim of this project falls under the broader *Urban Network Design Problem (UNDP)*, and more specifically within the *Discrete Network Design Problem (DNDP)*, which is generally considered NP-hard [1]. This complexity arises because network design involves bi-level decision-making: at the upper level, discrete infrastructure choices (such as reallocating or upgrading road segments) are made, while at the lower level, user flows adjust to the new network conditions. The interaction between these levels produces a non-convex and combinatorial search space, making it computationally intractable to find the global optimum for large-scale networks. In the literature, the upper-level optimization problem in bicycle network design is typically addressed through one of three main approaches: heuristic, linear or mixed-integer programming, and metaheuristic methods. At the lower level, models either rely on network-wide performance metrics (such as connectivity or efficiency) or on agent-based simulations that explicitly capture traveller behaviour. Each approach presents distinct advantages, limitations, and suitable use cases, balancing between computational efficiency, behavioural realism, and scalability. With this foundation on how these problems are conceptually structured, the following sections discuss the different modelling approaches in greater detail.

1.1 Segment Suggestion

Heuristic

The simplest of the methods used in bicycle network planning are heuristic approaches. These are considered “simple” because they rely on rule-based or greedy decision-making guided by intuitive network metrics such as connectedness, directness, or betweenness centrality (defined in [Evaluation](#)). Heuristic models iteratively add or remove edges in the bicycle network to improve a selected topological metric, without formulating the problem as a formal bi-level optimization task. In doing so, they make several simplifying assumptions, most notably, that traffic demand is uniformly distributed across the network and that travellers do not adjust their routes in response to infrastructure changes (i.e., no user equilibrium is modelled). As a result, these methods capture the geometric and structural characteristics of a network but neglect behavioural feedback and flow dynamics.

For instance [\[3\]](#) adopt a centrality-based heuristic, where new links are added according to graph-theoretic importance measures such as betweenness or closeness centrality. This approach prioritizes edges that are topologically critical or spatially central within the underlying street network, allowing the simulated bicycle network to grow along corridors that would most efficiently shorten travel distances.

In contrast, [\[4\]](#) propose component-based heuristics designed to reconnect fragmented pieces of existing infrastructure. Their *Largest-to-Second* (L2S) strategy links the two largest disconnected components to quickly consolidate the network, while the *Largest-to-Closest* (L2C) strategy connects the largest component to its nearest neighbour, favouring shorter and more spatially efficient connections. Both approaches share the same principle of incremental, topology-driven expansion, but differ in how they define the “importance” of the next edge to add: L2S biases towards overall connectedness, whereas L2C favours directness by preferring shorter spatial gaps. The authors also use simpler baselines for comparison, using *Random to Closest* (R2C), representing uncoordinated development, and *Closest Component* (CC), which connects the nearest two subgraphs, producing a minimum spanning tree that minimises investment but is not resilient to any interruptions. Overall, L2C performed best, with L2S also strong but slightly less effective, both outperforming the baselines. The study shows that targeted investments into key missing links can rapidly consolidate fragmented bicycle networks, though cars still achieve greater directness, with routes averaging about 33% shorter than those of bicycles.

A third paradigm is introduced by [\[5\]](#) applies percolation-based heuristics to model network growth and robustness over time. In a forward percolation approach, the network gradually expands by adding segments with the highest marginal contribution to global accessibility or welfare. Conversely, the inverse percolation approach begins from a fully built network and iteratively removes the least critical segments, recalculating connectivity and demand after each step. An additional advantage of the inverse percolation method is that it inherently maintains full network connectivity throughout the process, as segments are only removed if their deletion does not fragment the network. This ensures that all resulting configurations remain fully functional while progressively identifying the most critical corridors. These methods provide a transparent way to approximate how incremental investments or disinvestments affect overall network integrity.

Across heuristic models they consistently outperform baseline random growth strategies. Centrality-based growth yields higher connectedness, directness, and resilience than both random and minimum spanning tree baselines, producing networks that more closely resemble those of well-developed cycling cities such as Copenhagen [\[3\]](#). Component-based heuristics (L2S and L2C) outperform random or closest-component baselines by achieving faster consolidation and greater overall cohesion [\[5\]](#). These findings demonstrate that, even under uniform demand assumptions, simple topology-driven heuristics can reproduce high-quality, cohesive bicycle networks far more efficiently than unguided or piecemeal local planning.

While all heuristic families improve structural aspects of the network, they share fundamental simplifying assumptions. Each assumes uniform travel demand across the city and neglects behavioural factors such as route choice, mode preference, or induced demand. Centrality-based growth interprets betweenness as a proxy for flow under uniform conditions, component-based heuristics connect infrastructure purely by geometry, and percolation-based methods evaluate robustness statically without modelling user adaptation. Consequently, heuristic approaches provide a computationally efficient and transparent means of exploring large-scale bicycle networks and identifying missing or critical links, but their simplicity comes at the cost of behavioural realism, dynamic feedback, and global optimality.

Metaheuristic

Metaheuristic approaches aim to balance realism and computational feasibility, modelling the real world more accurately than simple heuristics while avoiding the exhaustive search required by exact Mixed-Integer Linear Programming (MILP) formulations. To achieve this, [6, 7] employ the Non-dominated Sorting Genetic Algorithm II (NSGA-II) to solve multi-objective optimization problems that simultaneously consider conflicting objectives for cyclists and car users. NSGA-II is an evolutionary algorithm that iteratively evolves a population of candidate solutions, ranking them based on their Pareto dominance, how well they perform relative to others across multiple objectives without being strictly better or worse in all of them. The outcome is a Pareto front, a set of optimal trade-off solutions where improving one objective (e.g., cyclist safety or travel time) would necessarily worsen another (e.g., car travel efficiency). Within this front, crowding distance is used to preserve diversity among non-dominated solutions, preventing premature convergence to a single region of the search space. After convergence, the final non-dominated set is further refined using the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method to rank and select the most balanced designs. This approach enables planners to explore a spectrum of balanced network configurations rather than a single, rigid optimum.

In [6] when evaluating the model on the large-scale case study on the historical-Cultural district of Tehran, Iran (approximately 2,000 hectares, 55 nodes, and 98 arcs), the proposed NSGA-II bi-level model demonstrated strong computational efficiency and scalability. The lower-level multi-class pedestrian assignment model (MPAM) was solved with an accuracy threshold of 10^3 , ensuring convergence while maintaining tractability for the large network. The optimization process required approximately 8,100 MPAM evaluations, compared to over 1,048,000 evaluations that would be needed under full enumeration, representing a significant reduction in computational effort. The test network included roughly 1,800 origin–destination (OD) pairs representing both work and leisure trips. The resulting Pareto front provided planners with a spectrum of trade-off solutions balancing travel cost, environmental quality, and construction cost, confirming that the NSGA-II approach can efficiently solve complex, behaviourally informed network design problems at a practical scale.

MILP

1.2 Evaluation

The second layer of the bi-level approach is a critical step, evaluating how much 'better' a suggested network. As stated before this is primarily solved using one of two methods, either using topographical network metrics or using agent based simulations. However before discussing these two methods we must first explore the different ways in which flow is represented.

Accurate flow estimation is fundamental to both modelling and evaluating transport networks, as it determines how travel demand is distributed across the available infrastructure. Two principal data sources are used for this purpose: Origin–Destination (OD) matrices and trajectory data. An OD matrix defines the total number of trips between each origin and destination pair within the study area, usually derived from household travel surveys or census statistics. OD data provides a macroscopic view of mobility patterns, making it suitable for equilibrium-based and aggregate network models where flow assignment depends on total demand rather than individual behaviour. For instance, [6, 7] both rely on OD data to simulate pedestrian and cycling flows at a city scale. Another instance, in the Tehran case study, [6] estimated the OD matrix using a combination of household surveys and field observations, such as counting pedestrians at subway stations and bus stops. Trips were classified into home-based and non-home-based categories, further subdivided into work and leisure purposes, allowing the model to distinguish between trip types with different sensitivities to distance and environmental factors. This approach ensured that the derived OD matrices reflected empirically observed travel behaviour, providing a realistic foundation for the lower-level equilibrium assignment. While [3] assume uniform OD demand between all node pairs to isolate the topological effects of network structure.

In contrast, trajectory data, such as GPS traces, or crowdsourced cycling data (e.g., Strava), captures individual-level travel behaviour with high spatial and temporal precision. This data enables the reconstruction of actual routes taken by users, revealing preferences for safety, directness, or comfort that OD matrices cannot capture. For example, [4] employ bike trajectory data to identify observed cyclist movement patterns and use these trajectories to guide the placement of new cycling links, thereby improving the behavioural realism of their model. Trajectory datasets are therefore particularly valuable for agent-based, where individual route choices and responses to infrastructure are

explicitly modelled. It is important to note that the trajectory data can be used to encode an OD matrix.

Each data source presents trade-offs. Trajectory data provides detailed behavioural information but often suffers from sampling bias (e.g., overrepresentation of fitness-oriented or app-using cyclists) and privacy limitations, while OD matrices are more systematically collected but lack route-level detail and temporal variation.

Metric

Agent

1.3 Methodology

References

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